ECG Heartbeat Classification

Theory Handbook

A Beginner-Friendly Illustrated Guide
to Deep Learning for ECG Arrhythmia Detection

Project: 1-D Residual CNN with SMOTE on MIT-BIH Dataset

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Target Audience: Students with basic programming knowledge, minimal DL background
Estimated Reading Time: 45-60 minutes
Purpose: Viva preparation, presentation support, concept review

Table of Contents

Chapter 1

Understanding ECGs and Heartbeats

1.1 How the Heart Works

Your heart is a muscular pump roughly the size of your fist. It beats about 100,000 times per day, pushing blood through your entire body. It has four chambers: two upper chambers called atria (singular: atrium) and two lower chambers called ventricles.

Think of it like a two-story house. The top floor (atria) receives blood, and the bottom floor (ventricles) pumps it out. The right side handles blood going to the lungs; the left side handles blood going to the rest of your body.

Every heartbeat follows a precise electrical sequence. This electrical activity is what an ECG measures.

1.2 The Electrical Conduction System

The heart has its own natural pacemaker -- a cluster of specialized cells that generate electrical signals. Here is the pathway that each electrical impulse follows:

Heart's Electrical Pathway

```
  SA Node (Sinoatrial Node)
    |  'The natural pacemaker'
    |  Located in the right atrium
    |  Fires ~60-100 times per minute
    v
AV Node (Atrioventricular Node)
    |  'The gatekeeper'
    |  Introduces a brief delay
    |  Allows atria to finish contracting
    v
Bundle of His
    |  'The highway'
    |  Carries signals into the ventricles
    |  Splits into left and right branches
    v
Purkinje Fibres
    |  'The delivery network'
    |  Distributes signal throughout ventricles
    v
Ventricular Contraction (the powerful pump)
```

Real-world analogy: Imagine a relay race. The SA node is the starting runner. It passes the baton to the AV node (who pauses briefly), then to the Bundle of His (who runs down a corridor), and finally to the Purkinje fibres (who spread out and trigger the finish).

If any part of this relay breaks down or fires at the wrong time, the result is an arrhythmia -- an irregular heartbeat.

Important:
The SA node is the heart's natural pacemaker. If it fails, the AV node takes over at a slower rate (~40-60 bpm). This backup system is why the heart is remarkably resilient.

1.3 What an ECG Records

An electrocardiogram (ECG or EKG) is a non-invasive test that records the electrical activity of the heart. Small electrode patches are placed on the skin. As the heart's electrical signals travel through the body, these electrodes detect tiny voltage changes and plot them as a waveform on paper or a screen.

Think of it like a seismograph for your heart. Just as a seismograph detects vibrations in the earth, an ECG detects electrical vibrations from your heart.

A standard clinical ECG uses 12 leads (different angles/perspectives of the heart). However, our project uses Lead II, which is the most common single-lead configuration. It measures the electrical difference between the right arm and left leg, providing an excellent view of the heart's primary electrical axis.

1.4 ECG Waveform Components

Every single heartbeat produces a characteristic waveform with distinct parts:

ECG Waveform Components

```
                        R
                     /|\
                    / | \
                   /  |  \
         P        /   |   \       T
       /   \     /    |    \    /   \
      /     \ /       |     \ /      \
-------        Q     |      S       --------
                      |
|--PR--||---QRS---||-----ST-----||---|
interval   complex    segment
```

P Wave

The P wave represents atrial depolarization -- the electrical signal spreading across the atria (upper chambers), causing them to contract and push blood into the ventricles. It is a small, rounded bump.

QRS Complex

The QRS complex represents ventricular depolarization -- the electrical signal spreading through the ventricles (lower chambers), causing the powerful contraction that pumps blood to the body. It is the tallest, sharpest part of the waveform. The R peak is the highest point and is used as the reference point for segmenting individual heartbeats.

T Wave

The T wave represents ventricular repolarization -- the ventricles resetting their electrical state in preparation for the next beat. It is a broader, gentler bump after the QRS complex.

Key Intervals

PR Interval: Time from the start of the P wave to the start of the QRS complex. Represents the delay at the AV node.

ST Segment: The flat line between the end of the QRS and the start of the T wave. Elevation or depression of this segment can indicate a heart attack.

QT Interval: From the start of the QRS to the end of the T wave. Represents the total time for ventricular contraction and recovery.

Remember:
In our project, each heartbeat is a window of 187 data points sampled at 125 Hz, centered around the R-peak. This captures approximately 1.5 seconds of ECG signal -- enough to see the P wave, QRS complex, and T wave.

1.5 What is an Arrhythmia?

An arrhythmia is any deviation from the normal, regular heartbeat rhythm. The word literally means 'without rhythm.' Arrhythmias can be:

- Too fast (tachycardia -- more than 100 bpm)
- Too slow (bradycardia -- less than 60 bpm)
- Irregular (beats coming at unpredictable times)
- Originating from the wrong location (e.g., ventricles firing on their own)

Some arrhythmias are harmless (many people experience occasional premature beats). Others can be life-threatening and require immediate medical intervention. The key challenge is telling them apart, which is exactly what our project automates.

1.6 The Five Heartbeat Classes

Our project classifies every heartbeat into one of five categories defined by the AAMI EC57 standard (Association for the Advancement of Medical Instrumentation):

Class	Label	Medical Name	Count	%	Clinical Risk
0	N	Normal / Bundle Branch Block	72,471	82.7%	Low
1	S	Supraventricular Ectopic	2,223	2.5%	Moderate
2	V	Ventricular Ectopic	5,788	6.6%	High
3	F	Fusion Beat	641	0.7%	Moderate
4	Q	Paced / Unclassifiable	6,431	7.3%	Variable

Class N -- Normal

These are healthy heartbeats following the normal conduction pathway. The SA node fires, the signal travels normally through the AV node and into the ventricles. The ECG shows a clean P wave, a narrow QRS complex, and a smooth T wave. This class also includes left and right bundle branch blocks (LBBB/RBBB), which are conduction delays that widen the QRS complex but are generally non-dangerous.

Class V -- Ventricular Ectopic

These beats originate in the ventricles instead of the SA node. Because the electrical signal travels through the ventricles in an abnormal path, the QRS complex is typically wide and bizarre-looking. Frequent ventricular ectopic beats can indicate serious heart disease and may lead to ventricular tachycardia or fibrillation (a life-threatening emergency).

Class Q -- Paced / Unclassifiable

These include beats generated by an artificial pacemaker device. They have a very distinctive spike before the QRS complex (the pacemaker stimulus artifact). This class also includes beats that could not be reliably classified by the annotators.

1.7 Deep Dive: Class S (Supraventricular Ectopic Beat)

Supraventricular ectopic beats originate above the ventricles -- specifically in the atria or the AV junction. The word 'supraventricular' literally means 'above the ventricles.'

Origin: Instead of the SA node initiating the beat, an abnormal focus somewhere in the atria or near the AV node fires prematurely. This is called a premature atrial contraction (PAC) or premature junctional contraction.

ECG Appearance: Because the signal still enters the ventricles through the normal conduction pathway (Bundle of His and Purkinje fibres), the QRS complex usually looks normal -- narrow and sharp, just like a Class N beat. The main differences are:

- The P wave may be absent, inverted, or buried within the QRS
- The timing is premature (it comes earlier than expected)
- The RR interval (time between consecutive R peaks) is irregular

Why It Is Hard to Classify: Class S beats look extremely similar to Class N beats. The QRS morphology is nearly identical. The subtle differences in the P wave and timing are the only distinguishing features, and these can be very small in the 187-point signal windows.

Clinical Importance: Frequent supraventricular ectopic beats can trigger supraventricular tachycardia (SVT), which causes rapid heartbeat, dizziness, and fainting. While rarely life-threatening on their own, they are important early warning signs.

Why Recall Matters: If the AI model misses a supraventricular ectopic beat (labels it as 'Normal'), the patient loses an early warning of potential SVT. In our baseline model, recall for Class S was only 79.3% -- meaning the model missed about 1 in 5 of these beats.

> **Viva Tip:**
> If asked 'Why is Class S hard to detect?', answer: Because the QRS complex looks almost identical to a normal beat. The only difference is a subtle change in the P wave and timing.

1.8 Deep Dive: Class F (Fusion Beat)

A fusion beat is one of the most fascinating and challenging phenomena in ECG analysis. It occurs when TWO electrical impulses activate the heart at the same time.

What Fuses: A normal impulse (from the SA node, traveling downward) and a ventricular impulse (originating in the ventricle, traveling upward) collide and simultaneously activate different parts of the ventricle. The resulting QRS complex is a 'hybrid' or 'blend' of a normal beat and a ventricular ectopic beat.

Why It Occurs: The timing must be extremely precise. The normal impulse must be descending at the exact moment a ventricular focus fires. This makes fusion beats inherently rare.

Why It Is Rare: Only 641 fusion beats exist in the training set (0.7%). This extreme rarity makes it very difficult for machine learning models to learn the pattern. The model simply does not see enough examples.

Why It Is Difficult to Classify: The ECG morphology of a fusion beat is somewhere between a normal beat and a ventricular beat. It is not quite normal and not quite ventricular -- it sits in a gray zone that confuses both human cardiologists and machine learning models.

Think of it like mixing two paint colors: the result is a new shade that does not clearly belong to either original color.

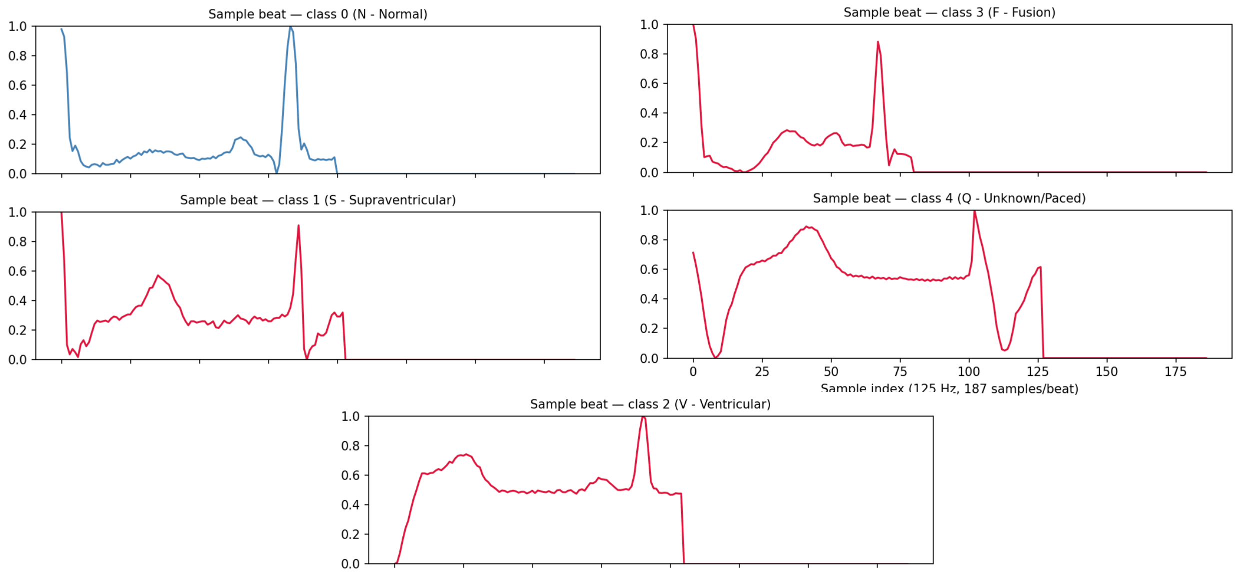
Clinical Importance: Fusion beats indicate that the ventricle has an independent electrical focus that is actively competing with the normal conduction pathway. This is a warning sign that the patient may be developing sustained ventricular arrhythmias, which can be life-threatening.

Why Improving F Detection Is Critical: In our baseline model, recall for Class F was only 78.4%. This means more than 1 in 5 fusion beats were being missed entirely. SMOTE improved this to 82.7% (+4.3%), which is a meaningful clinical improvement.

> **Common Mistake:**
> Do not confuse Fusion Beats with simply 'bad signal quality.' A fusion beat is a real, clinically significant phenomenon where two electrical impulses genuinely collide in the heart.

1.9 Visual Comparison of All Classes

The following figure shows sample ECG waveforms from each of the five classes:



Notice how Class N and Class S look very similar (both have narrow QRS complexes), while Class V has a wide, bizarre QRS. Class F appears as a blend between normal and ventricular morphology. Class Q shows the distinct pacemaker spike.

Chapter 2

Machine Learning Foundations

2.1 Datasets and the MIT-BIH Database

A dataset in machine learning is like a textbook for the computer. Just as a student learns from examples in a textbook, a machine learning model learns from examples in a dataset.

Our project uses the PhysioNet MIT-BIH Arrhythmia Database, which is the gold standard dataset for ECG heartbeat classification. It contains ECG recordings from 47 real patients.

The data has been preprocessed into two CSV files:
- Training set: 87,554 heartbeat samples (the textbook)
- Test set: 21,892 heartbeat samples (the final exam)

Each sample is a row of 188 numbers:
- Columns 0-186: The ECG signal (187 amplitude values, normalized between 0 and 1)
- Column 187: The class label (0, 1, 2, 3, or 4)

The signal represents approximately 1.5 seconds of ECG data, sampled at 125 Hz (125 measurements per second), centered around the R-peak of each heartbeat.

Important:
The dataset is already preprocessed. Each row is one heartbeat, already segmented, centered around the R-peak, and normalized. In real clinical applications, significant preprocessing (filtering, baseline correction, R-peak detection) would be needed before reaching this clean format.

2.2 Class Imbalance -- The Core Problem

Class imbalance is the single most important challenge in this project.

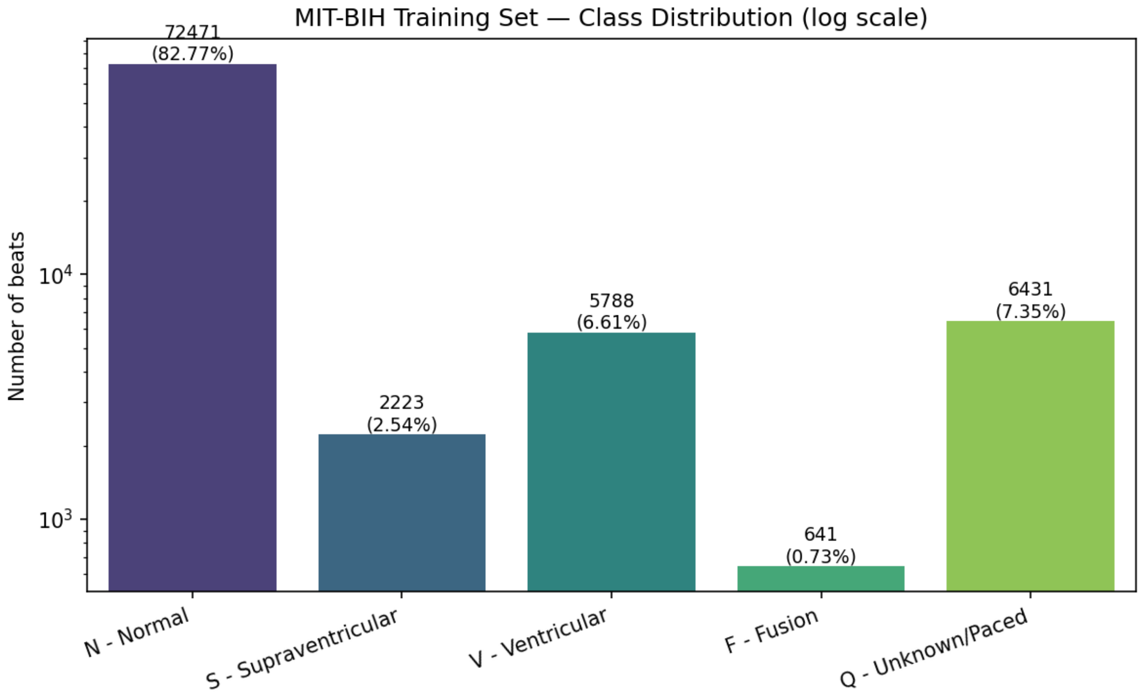
Imagine you are a teacher grading an exam where 83 out of 100 students wrote 'A', and only 1 student wrote 'F'. If you decide to just predict 'A' for every student without reading their answers, you would be correct 83% of the time. That sounds great, right? But you would completely miss the one student who needed help.

This is exactly what happens in our dataset:
- Normal beats (N): 82.7% of all data
- Fusion beats (F): Only 0.7% of all data

A lazy model that always predicts 'Normal' would achieve 82.7% accuracy while completely failing at its actual job: detecting dangerous arrhythmias.

This is why accuracy alone is a dangerously misleading metric in medical AI.

Here is the actual class distribution from our training data:



Common Mistake:
Never evaluate a medical AI model using only accuracy. Always check per-class recall (sensitivity) to ensure the model actually detects the rare, important conditions.

2.3 Core Machine Learning Concepts

Training vs Testing

Training data is like homework -- the model learns from it. Testing data is like a final exam -- the model has never seen it before, and its performance on this unseen data tells us how well it generalizes.

Critically: we never allow any information from the test set to influence training. This would be like giving students the exam answers before the test.

Features and Labels

Features are the inputs (the 187 ECG signal values). Labels are the correct answers (the class: 0, 1, 2, 3, or 4). The model's job is to learn the relationship between features and labels.

Prediction and Loss

A prediction is the model's guess for a given input. Loss is a number that measures how wrong the prediction is. During training, the model adjusts its internal parameters to minimize the loss.

Analogy: Imagine throwing darts at a target. The loss is the distance between where your dart lands and the bullseye. Each throw, you adjust your aim to get closer.

Overfitting and Generalization

Overfitting happens when the model memorizes the training data instead of learning the underlying patterns. It performs beautifully on training data but fails on new, unseen data.

Analogy: A student who memorizes every homework answer word-for-word but cannot solve a new problem on the exam. The student 'overfitted' to the homework.

Generalization is the opposite: the model learns the underlying rules and can apply them to new data it has never seen. This is the goal.

Tip:
EarlyStopping is a technique used in our project to prevent overfitting. It monitors the model's performance on a validation set and stops training when performance stops improving -- like a coach pulling a player off the field before they get too tired.

2.4 Neural Networks

A neural network is inspired by (but not identical to) the human brain. It consists of layers of small computational units called 'neurons.' Each neuron receives inputs, multiplies them by learned weights, adds them up, and passes the result through an activation function.

Analogy: Think of a neural network as a series of committees. Each committee member (neuron) receives information, forms an opinion (weighted sum), and votes (activation). The next committee takes these votes as input and forms higher-level opinions. The final committee makes the decision.

Key components:
- Input Layer: Receives the raw data (187 ECG values)
- Hidden Layers: Process and transform the data
- Output Layer: Produces the final prediction
- Weights: Learnable parameters that determine each neuron's behavior
- Activation Function: Introduces non-linearity (without it, the network could only learn linear patterns)

2.5 Convolutional Neural Networks (CNN)

A Convolutional Neural Network (CNN) is a special type of neural network designed to automatically detect patterns in structured data (images, signals, text).

The key innovation is the convolution operation: instead of connecting every input to every neuron (which is expensive and wasteful), a CNN slides a small 'filter' (also called a 'kernel') across the input, looking for local patterns.

Why 1D CNN Instead of 2D CNN?

2D CNNs are designed for images (which have two spatial dimensions: width and height). ECG signals are one-dimensional -- they only have one dimension: time. Using a 2D CNN on a 1D signal would be like using a metal detector designed for flat ground to search the ocean. It would work, but it would be wasteful, inefficient, and unnecessarily complex.

A 1D CNN slides its filter along one dimension (time), which is perfectly suited for detecting temporal patterns in ECG signals, such as the shape of the QRS complex or the timing between waves.

1D Convolution Visualized

```
  ECG Signal: [0.1, 0.3, 0.8, 1.0, 0.7, 0.2, 0.1, ...]
                 |___|___|___|___|___|
                 Filter (kernel) slides across

Filter: [w1, w2, w3, w4, w5]  (5 learnable weights)

At each position:
  output = w1*x1 + w2*x2 + w3*x3 + w4*x4 + w5*x5

The filter learns to detect specific shapes
(e.g., the sharp spike of a QRS complex)
```

Filters and Feature Maps

A filter is a small window of learnable weights. In our project, each convolutional layer uses 32 filters, each of size 5. This means the network learns 32 different pattern detectors, each looking at 5 consecutive time steps. The output of applying a filter across the signal is called a 'feature map.'

Pooling

Pooling reduces the size of the feature maps, keeping only the most important information. In our project, we use MaxPool1D with a pool size of 5 and stride of 2. This means: look at every 5 consecutive values, keep only the maximum, and move forward by 2 steps.

Analogy: Reading a long newspaper article and highlighting only the most important sentence in every paragraph. You reduce the length but keep the key information.

Flatten

After all convolutions and pooling, the data is a multi-dimensional array. The Flatten operation converts this into a single 1D vector so it can be fed into traditional Dense (fully connected) layers.

Dense (Fully Connected) Layers

Dense layers connect every neuron to every neuron in the previous layer. They act as the 'decision-making' part of the network, interpreting the features extracted by the CNN layers.

Softmax

The final layer uses softmax activation, which converts raw scores into a probability distribution across the 5 classes. The probabilities always sum to 1.0. The class with the highest probability is the prediction.

Example: [0.92, 0.02, 0.03, 0.01, 0.02] means the model is 92% confident this is a Normal beat.

2.6 Residual Networks (ResNets)

Residual Networks are a breakthrough architecture that solved one of the biggest problems in deep learning: the vanishing gradient problem.

The Vanishing Gradient Problem

When training a deep neural network, the model learns by sending error signals backward through the layers (this process is called backpropagation). However, as these signals pass through many layers, they get smaller and smaller -- like a whisper passed through a long chain of people. By the time it reaches the first layers, the signal is too weak to cause meaningful learning.

This means deeper networks (with more layers) can actually perform WORSE than shallower ones, even though they have more capacity. This was a major paradox in deep learning before ResNets.

Skip Connections: The Solution

ResNets solve this with a brilliantly simple idea: skip connections (also called shortcut connections). Instead of forcing the signal to pass through every layer, the network adds a 'shortcut' that bypasses some layers and adds the original input directly to the output.

Analogy: Imagine a highway with a series of toll booths (layers). A skip connection is like adding an express lane that bypasses the toll booths. The traffic (gradient signal) can take either the regular lanes or the express lane, ensuring it always reaches the destination.

Residual Block: Skip Connection

```
  Input ----+----> [Conv1D] ----> [ReLU] ----> [Conv1D] ----+----> [Add] ----> [ReLU]
            |                                               |
            +-------------- Skip Connection ----------------+
                        (identity shortcut)

The output = F(input) + input
The network only needs to learn the RESIDUAL: F(input)
If the layers learn nothing useful, F(input) = 0,
and the output simply equals the input (no harm done).
```

This is why it is called 'residual' learning. Instead of learning the entire transformation, the network only needs to learn the difference (residual) between the input and the desired output. If the extra layers do not help, the network can simply learn to set their contribution to zero, and the data passes through unchanged.

Viva Tip:
If asked 'What is the key advantage of ResNets?', answer: Skip connections allow the gradient to flow directly through the network, solving the vanishing gradient problem and enabling training of much deeper networks without degradation.

Chapter 3

Our Proposed Model

3.1 Project Pipeline

Our project follows a structured experimental pipeline:

Complete Project Pipeline

```
  [MIT-BIH Dataset]
      |
      v
[Data Loading & Preprocessing]
CSV -> Numpy Array -> Reshape to (N, 187, 1)
      |
      +------------------+
      |                  |
      v                  v
[BASELINE PATH]    [SMOTE PATH]
Use raw data       Flatten -> SMOTE -> Reshape
      |                  |
      v                  v
[Train 1-D ResNet] [Train 1-D ResNet]
(identical CNN)    (identical CNN)
      |                  |
      v                  v
[Evaluate on       [Evaluate on
 SAME test set]     SAME test set]
      |                  |
      +------------------+
      |
      v
[Compare Results]
Confusion Matrices, F1 Scores, Recall
```

The key principle: CONTROLLED EXPERIMENT. Both models use the exact same CNN architecture, the exact same hyperparameters, and are evaluated on the exact same test set. The ONLY difference is whether the training data was balanced using SMOTE. This isolates SMOTE as the single variable being tested.

3.2 SMOTE: Synthetic Minority Over-sampling Technique

Why Not Just Copy Minority Samples?

The simplest approach to class imbalance is random oversampling: make copies of the minority class samples until all classes have equal counts. However, this causes severe overfitting because the model sees the exact same samples multiple times and memorizes them rather than learning generalizable patterns.

Analogy: If a student has only one practice problem for a difficult topic, copying that same problem 100 times does not help them learn. They just memorize the specific answer.

How SMOTE Works

SMOTE creates brand-new synthetic samples by interpolating between existing minority class samples:

Step 1: Pick a minority sample (let's call it A)
Step 2: Find its k nearest neighbors (default k=5) among other samples of the same class
Step 3: Randomly pick one of those neighbors (call it B)
Step 4: Create a new synthetic sample at a random point on the line between A and B

Mathematically: new_sample = A + random(0,1) * (B - A)

This creates a new sample that shares characteristics with both A and B but is not identical to either.

SMOTE Visualization (Simplified 2D)

```
  Before SMOTE:                   After SMOTE:

   O   O                        O x O
     O                          x O x
   O                            O x

O = original minority           x = synthetic samples
samples (few)                   (created between existing)

The synthetic samples fill in the gaps between
existing samples, expanding the decision boundary.
```

SMOTE in Our Project

In our project, each heartbeat is represented as 187 numbers (the ECG amplitude values). SMOTE treats these as coordinates in a 187-dimensional space and performs interpolation in that space.

Important technical detail: SMOTE requires 2D input (samples, features) but our CNN requires 3D input (samples, time_steps, channels). So we must:

1. Flatten from (87554, 187, 1) to (87554, 187)
2. Apply SMOTE
3. Reshape back to (N, 187, 1)

After SMOTE, all five classes have equal counts (each class has 72,471 samples).

Common Mistake:
SMOTE must ONLY be applied to the training data. If synthetic samples leak into the test set, the evaluation becomes meaningless because you are testing on artificially generated data that does not represent real patients.

SMOTE Limitations

SMOTE is not perfect:
- It performs linear interpolation, which assumes a straight-line path between two beats is biologically valid. This is not guaranteed.
- A synthetic ECG waveform created by SMOTE may not correspond to any real physiological heartbeat.
- It can create noisy or ambiguous samples near class boundaries.

Despite these limitations, SMOTE is widely used and validated in the ECG classification literature, and our results confirm it provides meaningful improvement.

3.3 CNN Architecture

Our model is a 1-D Residual CNN inspired by the architecture proposed by Kachuee et al. (2018).

Complete Model Architecture

```
  Layer                          Output Shape
------------------------------------------
Input                          (187, 1)
Conv1D(32, 5, same) + ReLU     (187, 32)

Residual Block 1               (94, 32)
Residual Block 2               (47, 32)
Residual Block 3               (24, 32)
Residual Block 4               (12, 32)
Residual Block 5               (6, 32)

Flatten                        (192,)
Dense(32) + ReLU               (32,)
Dense(32) + ReLU               (32,)
Dense(5) + Softmax             (5,)

Each Residual Block:
  Conv1D(32,5) -> ReLU -> Conv1D(32,5)
  -> Add(skip) -> ReLU -> MaxPool1D(5,2)
```

Key observations:
- 11 convolutional layers total (1 initial + 2 per block x 5 blocks)
- 5 residual blocks with skip connections
- 3 dense layers (2 hidden + 1 output)
- Input: 187 time steps with 1 channel
- Output: 5 probabilities (one per class)
- Each residual block halves the temporal resolution via MaxPool1D(5, stride=2)

3.4 Hyperparameters

Hyperparameters are settings chosen by the engineer BEFORE training begins. They are not learned by the model.

Parameter	Value	Why This Value?
Optimizer	Adam	Adapts learning rate per-parameter, fast convergence
Learning Rate	0.001	Standard starting point, stable for deep CNNs
Loss Function	Sparse Cat. CE	Handles integer labels directly, saves memory
Batch Size	256	Large enough for stable gradients with imbalanced data
Epochs	30	Sufficient with EarlyStopping to find optimum
Val. Split	10%	~8,755 samples for validation signal
EarlyStopping	patience=5	Stops when val loss stalls for 5 epochs
ReduceLR	patience=3	Halves LR when val loss stalls for 3 epochs

EarlyStopping Explained

EarlyStopping monitors the validation loss after every epoch. If the loss does not improve for a specified number of epochs (patience=5), training is stopped automatically. It also restores the model weights from the best epoch.

Analogy: Like a coach who notices a player is getting tired and keeps making mistakes. Instead of letting them play until the end, the coach pulls them off at their peak performance.

ReduceLROnPlateau Explained

When the validation loss stops improving for 3 consecutive epochs, this callback multiplies the learning rate by 0.5 (halves it). A smaller learning rate makes the optimizer take smaller steps, which helps it find better solutions when it is close to a minimum.

Analogy: When you are driving and you see your destination ahead, you slow down to park precisely. If you kept driving at highway speed, you would overshoot.

3.5 Design Decisions and Rationale

Every design choice has a specific reason:

Decision	Rationale
Conv1D not 2D	ECG is 1D time-series; 2D adds unnecessary complexity
ResNet not shallow CNN	Skip connections solve vanishing gradients for 13 layers
Adam not SGD	Adaptive LR converges faster on complex loss surfaces
SMOTE not Random Oversample	Creates novel samples instead of duplicates
Same architecture both exp.	Isolates SMOTE as the only variable (fair test)
Batch size 256	Stabilizes gradients in imbalanced distribution
Sparse CE not CE	Handles integer labels directly, less memory
10% validation split	8,755 samples = statistically significant signal

Chapter 4

Understanding the Experimental Results

4.1 Evaluation Metrics Explained

Before looking at results, you must understand what each metric means. These metrics appear in every machine learning project, so understanding them is essential.

Accuracy

Accuracy = (Number of correct predictions) / (Total predictions)

It answers: 'Overall, how often was the model right?'

Problem: With our imbalanced dataset, a model that always predicts 'Normal' would achieve 82.7% accuracy while being completely useless for detecting arrhythmias.

Precision

Precision = True Positives / (True Positives + False Positives)

It answers: 'When the model says a beat is Class X, how often is it correct?'

Analogy: If a spam filter marks 100 emails as spam, precision tells you how many of those 100 were actually spam. High precision = few false alarms.

Recall (Sensitivity)

Recall = True Positives / (True Positives + False Negatives)

It answers: 'Out of all actual Class X beats, how many did the model find?'

Analogy: If there are 100 actual spam emails in your inbox, recall tells you how many the filter caught. High recall = few missed cases.

Important:
In medical diagnosis, recall is usually MORE important than precision. Missing a dangerous arrhythmia (low recall) can cost a life. A false alarm (low precision) only costs a doctor a few minutes to double-check.

F1-Score

F1 = 2 * (Precision * Recall) / (Precision + Recall)

It is the harmonic mean of precision and recall, providing a single number that balances both. An F1 of 1.0 is perfect; an F1 of 0.0 means complete failure.

Support

Support is simply the number of actual instances of each class in the test set. For example, Class N has 18,118 test samples while Class F has only 162.

Macro Average vs Weighted Average

Macro average: Calculate the metric for each class, then take the simple average. Every class counts equally.

Weighted average: Same calculation, but each class is weighted by its support (number of samples). Classes with more samples have more influence.

In imbalanced datasets, macro average better reveals problems with minority classes because it does not let the majority class dominate the score.

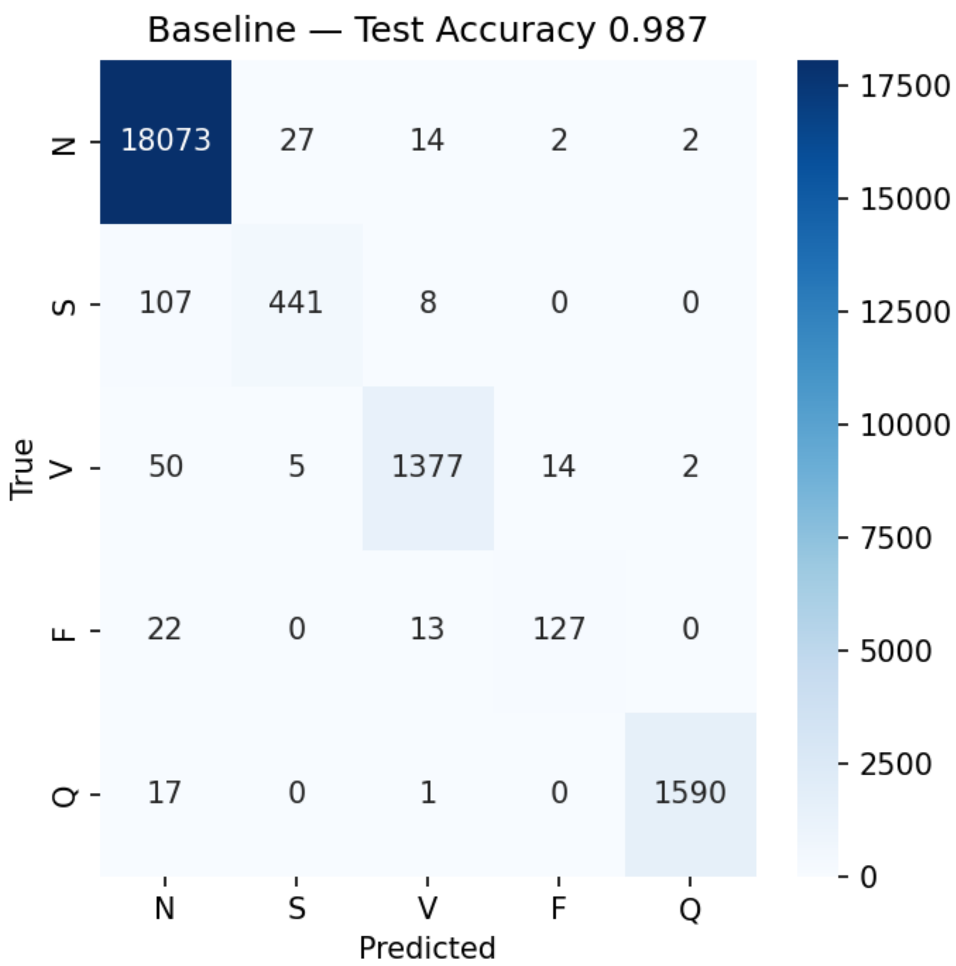
Confusion Matrix

A confusion matrix is a table showing exactly what the model predicted vs what the correct answer was. Each row represents the actual class, and each column represents the predicted class. The diagonal shows correct predictions; off-diagonal cells show errors.

It answers: 'When the model made a mistake, WHAT did it confuse the beat with?'

4.2 Baseline Results

The baseline model was trained on the raw, imbalanced dataset (no SMOTE).



Class	Precision	Recall	F1-Score	Support

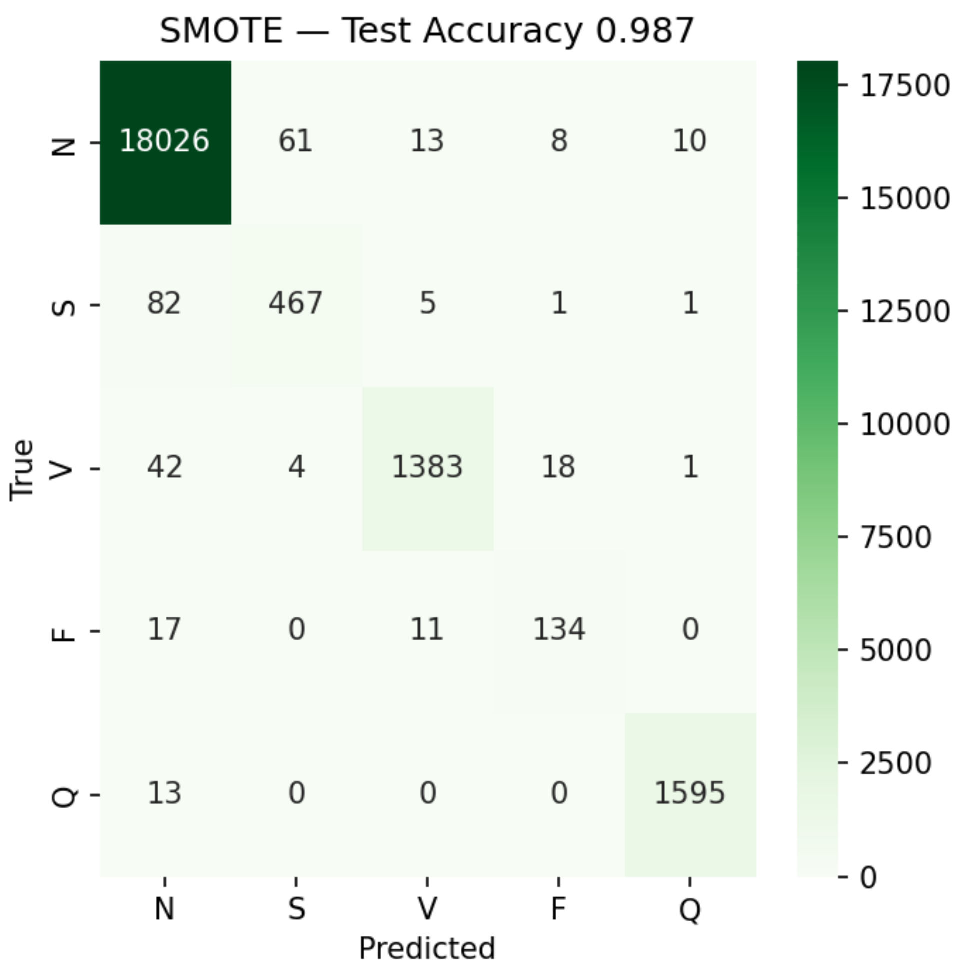
N (Normal)	98.93%	99.75%	99.34%	18,118
S (Supraventricular)	93.23%	79.32%	85.71%	556
V (Ventricular)	97.45%	95.10%	96.26%	1,448
F (Fusion)	88.81%	78.40%	83.28%	162
Q (Paced)	99.75%	98.88%	99.31%	1,608

Overall accuracy: 98.70%
Macro F1: 92.78%

Key observation: While overall accuracy is excellent (98.70%), recall for Class S (79.3%) and Class F (78.4%) is concerning. The model misses approximately 1 in 5 of these dangerous arrhythmias.

4.3 SMOTE Results

The SMOTE model was trained on the balanced dataset.



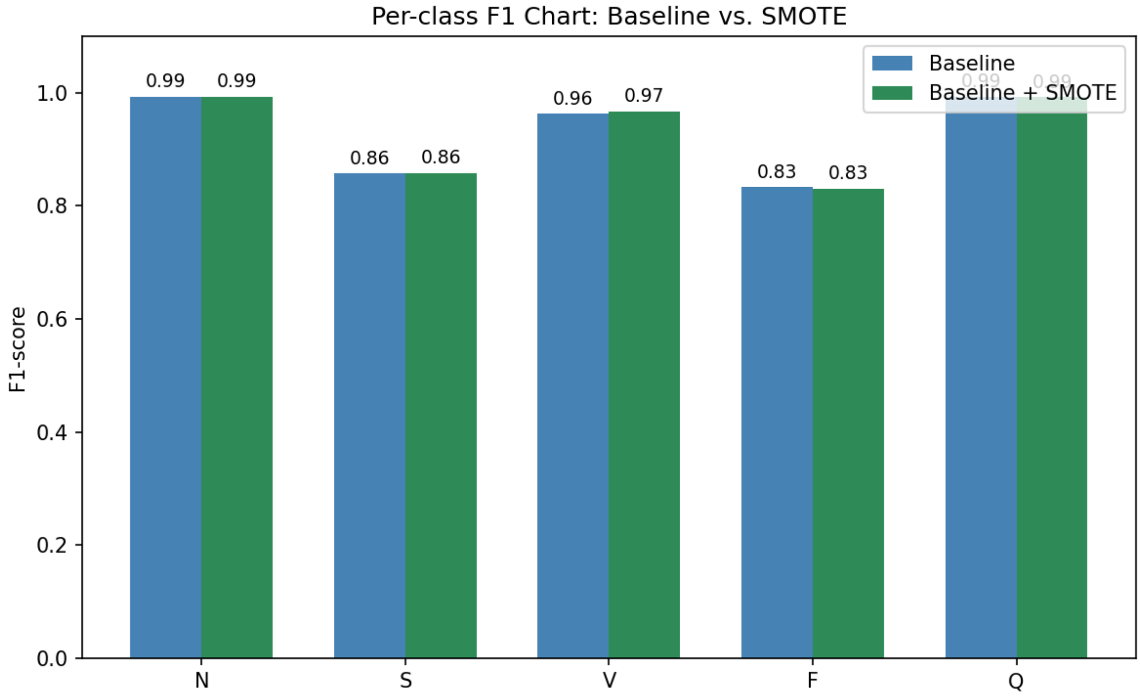
Class	Precision	Recall	F1-Score	Support
N (Normal)	99.15%	99.49%	99.32%	18,118
S (Supraventricular)	87.78%	84.00%	85.85%	556
V (Ventricular)	97.95%	95.51%	96.71%	1,448
F (Fusion)	83.23%	82.72%	82.97%	162

Q (Paced)	99.25%	99.19%	99.22%	1,608

Overall accuracy: 98.69%

Macro F1: 92.81%

4.4 Head-to-Head Comparison



Metric	Baseline	SMOTE	Change
Overall Accuracy	98.70%	98.69%	-0.01%
Macro F1	92.78%	92.81%	+0.03%
Recall (S)	79.32%	84.00%	+4.67%
Recall (F)	78.40%	82.72%	+4.32%
Recall (V)	95.10%	95.51%	+0.41%
Recall (Q)	98.88%	99.19%	+0.31%
Precision (S)	93.23%	87.78%	-5.45%
Precision (F)	88.81%	83.23%	-5.58%

4.5 The Precision-Recall Trade-off

The results clearly show a precision-recall trade-off:

SMOTE increased recall (the model catches more arrhythmias) but decreased precision (more false alarms).

Why? SMOTE gave the model more examples of minority classes, making it more 'suspicious' -- more likely to classify ambiguous beats as arrhythmias. This means:

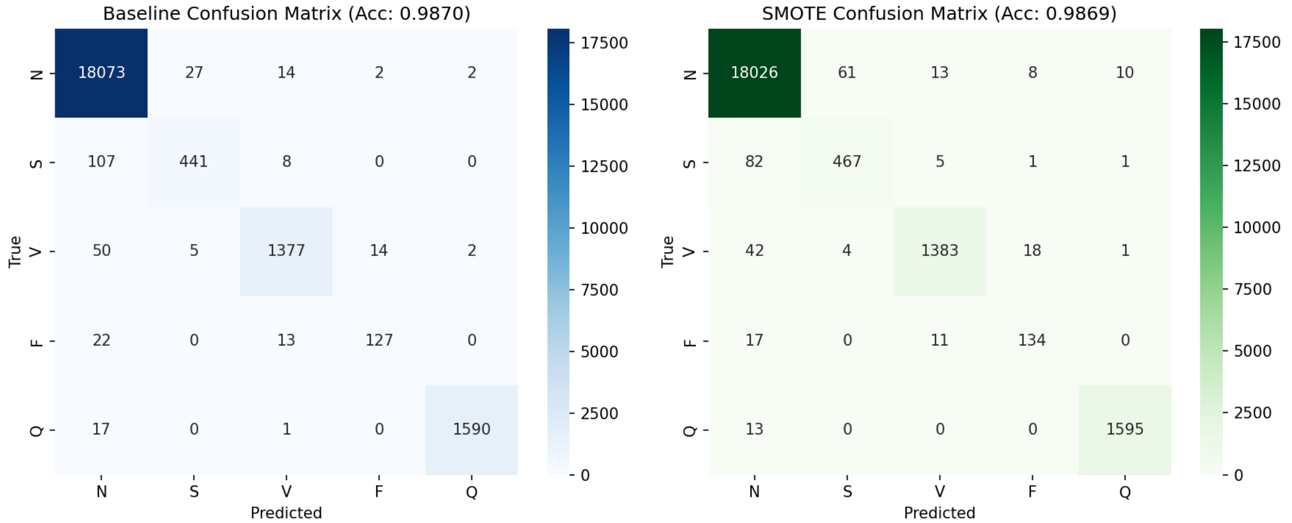
- More true arrhythmias are caught (higher recall, good!)
- But some normal beats are incorrectly flagged (lower precision, acceptable)

In medical context:

- A false positive means the doctor reviews a normal beat -- costs a few seconds
- A false negative means a dangerous arrhythmia is missed -- potentially fatal

Therefore, the SMOTE model is clinically superior despite having marginally lower accuracy.

Remember:
Overall accuracy stayed virtually flat (98.70% vs 98.69%) because Class N dominates the test set. The tiny decrease in Class N recall is invisible in the overall accuracy number, masking the substantial +4.67% improvement in Class S recall.



Chapter 5

Frequently Asked Questions (Viva Preparation)

This chapter contains 50 questions you are likely to encounter during a viva, interview, or technical discussion about this project. Each answer is written in simple, conversational language.

Q: Why did you choose a CNN for this task?

A: CNNs automatically learn relevant features from raw data through convolutional filters. For ECG classification, this means the network discovers patterns like QRS complex shape and timing without manual feature engineering.

Q: Why 1D CNN and not 2D CNN?

A: ECG signals are one-dimensional time series. Using a 2D CNN would require converting the signal into a 2D representation (like a spectrogram), adding complexity without benefit. A 1D CNN naturally processes temporal sequences.

Q: Why not use an LSTM or RNN?

A: LSTMs process sequences step-by-step, which is slower. Our signal windows are short (187 points), so the long-range memory advantage of LSTMs is unnecessary. CNNs can process the entire window in parallel and are more computationally efficient for fixed-length inputs.

Q: Why not use a Transformer?

A: Transformers excel with long sequences and large datasets. Our signals are only 187 points long, and our dataset, while substantial, may not fully benefit from the massive parameter counts of Transformers. The CNN provides an excellent accuracy-to-complexity ratio.

Q: What is a Residual Block?

A: A residual block contains two convolutional layers with a skip connection that adds the input directly to the output. This allows gradients to flow through the skip connection during backpropagation, preventing the vanishing gradient problem.

Q: Why use skip connections?

A: Without skip connections, deep networks suffer from vanishing gradients -- the error signal weakens as it travels backward through many layers. Skip connections provide a direct path for the gradient, enabling training of deeper networks without degradation.

Q: What is SMOTE?

A: SMOTE stands for Synthetic Minority Over-sampling Technique. It creates new synthetic examples of minority classes by interpolating between existing samples. This balances the class distribution without simply copying existing data.

Q: Why not use Random Oversampling instead of SMOTE?

A: Random oversampling duplicates existing minority samples, which causes the model to memorize those specific examples (overfitting). SMOTE creates new samples that share characteristics with existing ones but are not identical, promoting better generalization.

Q: Why is class imbalance a problem?

A: With 82.7% Normal beats, a model can achieve high accuracy by simply predicting Normal for everything. This means it never learns to detect rare arrhythmias, which is the actual clinical purpose of the system.

Q: Why is recall more important than accuracy?

A: In medical diagnosis, a missed arrhythmia (false negative) can be fatal. A false alarm (false positive) only requires a brief verification by a doctor. Recall measures how many actual arrhythmias were detected, making it the priority metric.

Q: Why is accuracy misleading in this project?

A: Because 82.7% of the test set is Class N. A 5% improvement in minority class recall is completely hidden by the overwhelming number of correctly classified normal beats. Accuracy stayed at 98.7% while recall for Class S improved by nearly 5%.

Q: What is a Fusion Beat (Class F)?

A: A fusion beat occurs when two electrical impulses (one normal from the SA node, one ectopic from the ventricle) simultaneously activate the heart. The resulting QRS complex is a blend of normal and ventricular morphology.

Q: Why is Class F difficult to classify?

A: Because fusion beats exist on a morphological spectrum between normal and ventricular beats. They do not clearly belong to either category, making them inherently ambiguous for both humans and machines.

Q: What is a Supraventricular Ectopic Beat (Class S)?

A: A beat originating above the ventricles (in the atria or AV junction) that fires prematurely. The QRS complex looks nearly identical to a normal beat because the ventricular conduction pathway is still used.

Q: Why is Class S difficult to classify?

A: Because the QRS complex is nearly identical to Class N. The only distinguishing features are subtle P-wave changes and timing irregularities, which may be difficult to detect in 187-point signal windows.

Q: What dataset did you use?

A: The MIT-BIH Arrhythmia Database from PhysioNet, preprocessed by Kachuee et al. Available on Kaggle. It contains 87,554 training and 21,892 testing samples across 5 AAMI classes.

Q: How are the ECG signals preprocessed?

A: Each heartbeat is a 187-point window sampled at 125 Hz, centered around the R-peak, and normalized to values between 0 and 1. We reshape from (N, 187) to (N, 187, 1) for the Conv1D layer.

Q: Why is the signal 187 points long?

A: At 125 Hz sampling rate, 187 points represent approximately 1.5 seconds of ECG. This window is long enough to capture the complete P-QRS-T complex of a single heartbeat.

Q: What is the role of the Flatten layer?

A: Flatten converts the multi-dimensional output of the last convolutional block into a single 1D vector, which is required as input for the Dense (fully connected) classification layers.

Q: What does the Softmax function do?

A: Softmax converts raw output scores into a probability distribution across the 5 classes. Each output is between 0 and 1, and all five values sum to exactly 1. The class with the highest probability is the prediction.

Q: Why did you use the Adam optimizer?

A: Adam combines the benefits of AdaGrad and RMSProp, providing adaptive learning rates for each parameter. It converges faster than standard SGD and is more robust to saddle points in the loss landscape.

Q: What is EarlyStopping?

A: A training callback that monitors validation loss. If the loss does not improve for a specified number of epochs (patience=5), training stops automatically. This prevents overfitting and ensures the model is saved at its optimal point.

Q: What is ReduceLROnPlateau?

A: A callback that reduces the learning rate by a factor (0.5) when validation loss stops improving for 3 epochs. Smaller learning rates help the optimizer make finer adjustments near the optimal solution.

Q: Why use Sparse Categorical Crossentropy?

A: Because our labels are integers (0-4) rather than one-hot encoded vectors. Sparse Categorical Crossentropy handles integer labels directly, saving memory and computation.

Q: Why is the batch size 256?

A: A larger batch size provides more stable gradient estimates per update step, which is important when the dataset is highly imbalanced. It also maximizes GPU utilization for efficient training.

Q: How many layers does your model have?

A: 11 convolutional layers (1 initial + 2 per residual block x 5 blocks), plus 3 dense layers (2 hidden + 1 output), totaling 14 learnable layers.

Q: How many residual blocks are there?

A: Five residual blocks, each containing two Conv1D layers, a skip connection with Add, ReLU activation, and MaxPool1D.

Q: What is the kernel size and why 5?

A: Kernel size 5 means each filter looks at 5 consecutive time steps. At 125 Hz, this covers 40ms of the signal, which is approximately the width of significant ECG features like the QRS complex.

Q: What happens inside a residual block?

A: Input passes through Conv1D -> ReLU -> Conv1D, then the original input (shortcut) is added to this output. The sum passes through ReLU and MaxPool1D (pool=5, stride=2), which halves the temporal dimension.

Q: Why did both models achieve almost the same accuracy?

A: Because 82.7% of the test set is Class N, which both models classify nearly perfectly. The accuracy metric is dominated by the majority class and is insensitive to changes in minority class performance.

Q: What improved with SMOTE?

A: Recall for Class S improved from 79.3% to 84.0% (+4.67%), and recall for Class F improved from 78.4% to 82.7% (+4.32%). The model catches significantly more rare arrhythmias.

Q: What got worse with SMOTE?

A: Precision for Class S dropped from 93.2% to 87.8% (-5.4%), and precision for Class F dropped from 88.8% to 83.2% (-5.6%). More false positives occur for these classes.

Q: Is the trade-off worth it?

A: Yes, in medical contexts. Catching 5% more dangerous arrhythmias at the cost of more false alarms is clinically favorable. A false alarm leads to a quick doctor review; a missed arrhythmia can be fatal.

Q: What is the confusion matrix?

A: A table where rows represent actual classes and columns represent predicted classes. Diagonal cells show correct predictions. Off-diagonal cells show what the model confused each class with.

Q: What is F1-Score?

A: The harmonic mean of precision and recall: F1 = 2*P*R/(P+R). It balances both metrics into a single number. It is useful when you need a single metric that accounts for both false positives and false negatives.

Q: What is Macro F1 vs Weighted F1?

A: Macro F1 averages F1 across all classes equally. Weighted F1 averages F1 weighted by class support. Macro F1 is more informative for imbalanced datasets because it gives equal importance to minority classes.

Q: Could you use data augmentation instead of SMOTE?

A: Yes. Techniques like time-warping, jittering, or window cropping could augment minority classes while maintaining temporal coherence. GANs could also generate realistic synthetic ECGs. These are future work.

Q: What is the difference between overfitting and underfitting?

A: Overfitting: model memorizes training data but fails on new data (too complex). Underfitting: model is too simple to capture the underlying patterns (performs poorly on both training and test data).

Q: Why not use class weights instead of SMOTE?

A: Class weighting modifies the loss function to penalize minority class errors more heavily. It is a valid alternative. SMOTE was chosen because it physically changes the data distribution, giving the model more diverse minority examples to learn from.

Q: What is the validation split used for?

A: 10% of the training data is held out as a validation set. It is used to monitor overfitting during training (via EarlyStopping and ReduceLROnPlateau) without touching the test set.

Q: Why not apply SMOTE to the test set?

A: Applying SMOTE to the test set would introduce synthetic data into the evaluation, making the results meaningless. The test set must represent real-world data distribution to provide an honest assessment.

Q: What framework was used?

A: TensorFlow with Keras API for model building and training. Scikit-learn for evaluation metrics. Imbalanced-learn for SMOTE implementation.

Q: How reproducible is this project?

A: Highly reproducible. SMOTE uses random_state=42 for consistent synthetic generation. The dataset is publicly available. The code is modular with separate files for data loading and model definition.

Q: What are the limitations of SMOTE on ECG data?

A: SMOTE interpolates linearly in feature space. A point between two ECG waveforms may not represent a physiologically valid heartbeat. The synthetic signals are mathematically plausible but not necessarily biologically realistic.

Q: What is the AAMI EC57 standard?

A: A medical standard that defines how heartbeat types should be grouped into 5 categories (N, S, V, F, Q) for automated classification. It ensures consistent labeling across different research studies.

Q: What is Lead II?

A: An ECG configuration measuring voltage between the right arm and left leg electrodes. It provides the best view of the heart's main electrical axis and is the most commonly used single lead.

Q: How would you deploy this model?

A: The saved .keras model file could be loaded by a FastAPI or TensorFlow Serving backend. Incoming ECG segments would be preprocessed to (1, 187, 1) format and passed to model.predict() for real-time classification.

Q: What would you improve if you had more time?

A: 1) Add Focal Loss to handle imbalance at the loss level. 2) Try Transformers for long-range dependencies. 3) Add Grad-CAM for explainability. 4) Test on external datasets for generalizability. 5) Implement patient-wise cross-validation.

Q: What is the clinical significance of this project?

A: Automated ECG classification can assist cardiologists in continuous monitoring scenarios (e.g., Holter monitors), flagging suspicious beats for review. It reduces the time and human error in analyzing thousands of heartbeats.

Q: Why is this better than a simple threshold-based approach?

A: Threshold-based methods require handcrafted features and work poorly with subtle morphological differences. Deep CNNs automatically learn the most discriminative features from raw data, achieving much higher accuracy on complex classification tasks.

Chapter 6

Common Misconceptions

"High accuracy means the model is good."

FALSE. In imbalanced datasets, accuracy is dominated by the majority class. A model that always predicts 'Normal' achieves 82.7% accuracy while being completely useless for detecting arrhythmias. Always check per-class recall.

"SMOTE creates fake patients."

FALSE. SMOTE creates synthetic ECG signal patterns, not fake patient records. Each synthetic sample is a mathematical interpolation between two real signals of the same class. It helps the model see more variation in minority classes.

"Precision and recall are the same thing."

FALSE. Precision asks: 'Of the ones I predicted as X, how many are actually X?' Recall asks: 'Of all actual X cases, how many did I find?' They measure fundamentally different aspects of model performance.

"CNNs understand ECGs like doctors do."

FALSE. CNNs are sophisticated pattern matchers. They detect statistical correlations between signal shapes and labels. They have no understanding of cardiac physiology, hemodynamics, or patient context. They are tools that assist doctors, not replace them.

"More training data always leads to better results."

PARTIALLY FALSE. More data generally helps, but if the additional data is heavily skewed toward one class, it can actually make the imbalance problem worse. Quality and balance of data matter as much as quantity.

"A deeper network is always better."

FALSE. Without techniques like skip connections, deeper networks suffer from vanishing gradients and actually perform worse. ResNets solved this specific problem, but even with skip connections, unnecessarily deep networks waste computation and can overfit.

"SMOTE guarantees better results."

FALSE. SMOTE typically improves minority class recall but may decrease precision. The net effect depends on the specific dataset and model. In our project, it provided a clear clinical improvement, but this is not universal.

"The test set should also be balanced."

FALSE. The test set should represent the real-world distribution. If the real world has 82.7% normal beats, the test set should too. Balancing the test set would give an unrealistic picture of how the model performs in practice.

"99% accuracy means the model is nearly perfect."

FALSE. In a dataset where 99% of samples belong to one class, a trivial model that always predicts that class achieves 99% accuracy. The remaining 1% (which may be the most important cases) is completely ignored.

"Neural networks are black boxes that cannot be understood."

PARTIALLY FALSE. While interpreting individual neurons is difficult, techniques like Grad-CAM, attention maps, SHAP, and LIME can highlight which parts of the input the model focuses on. In ECG classification, these tools can show whether the model is attending to the QRS complex or P wave.

Chapter 7

Glossary

AAMI	Association for the Advancement of Medical Instrumentation. Defines the EC57 standard for grouping heartbeat types into 5 categories.
Accuracy	The proportion of correct predictions out of all predictions. Misleading in imbalanced datasets.
Activation Function	A mathematical function applied after a neuron's weighted sum. Introduces non-linearity. Common examples: ReLU, Sigmoid, Softmax.
Adam	Adaptive Moment Estimation. An optimizer that maintains per-parameter learning rates using first and second moment estimates of the gradient.
Arrhythmia	Any heartbeat rhythm that deviates from the normal sinus rhythm. Can be fast, slow, or irregular.
AV Node	Atrioventricular Node. The electrical relay station between the atria and ventricles. Introduces a brief delay.
Backpropagation	The algorithm used to calculate how each weight contributes to the error, enabling the model to learn by adjusting weights.
Batch Size	The number of training samples processed before the model updates its weights. Larger batches give more stable gradients.
CNN	Convolutional Neural Network. A neural network architecture using convolutional filters to detect local patterns.
Confusion Matrix	A table showing predicted vs actual class labels. Diagonal = correct, off-diagonal = errors.
Conv1D	1-Dimensional Convolution. Slides a filter along one axis (time) to detect temporal patterns in sequential data.
Dense Layer	A fully connected layer where every neuron connects to every neuron in the previous layer.
EarlyStopping	A callback that halts training when validation performance stops improving, preventing overfitting.
ECG	Electrocardiogram. A test recording the heart's electrical activity via skin electrodes.
Epoch	One complete pass through the entire training dataset.
F1-Score	Harmonic mean of precision and recall. F1 = 2*P*R/(P+R). Balances both metrics.
Feature Map	The output of a convolutional filter applied across the input. Highlights where specific patterns appear.
Flatten	Operation that converts multi-dimensional data into a single 1D vector.
Fusion Beat	A heartbeat created when a normal and ventricular impulse simultaneously activate the heart.
Generalization	The ability of a model to perform well on unseen data, not just the training data.

Gradient	The direction and magnitude of steepest descent in the loss function. Used to update model weights.
Hyperparameter	A setting chosen before training (e.g., learning rate, batch size). Not learned by the model.
Imbalanced Dataset	A dataset where some classes have far more samples than others.
Kernel	Another name for a convolutional filter. A small window of learnable weights.
Lead II	An ECG configuration measuring voltage between right arm and left leg electrodes.
Loss Function	A function that measures how wrong the model's predictions are. Training aims to minimize it.
MaxPooling	A downsampling operation that keeps only the maximum value in each window, reducing data size.
MIT-BIH	Massachusetts Institute of Technology - Beth Israel Hospital. The source of the gold-standard arrhythmia database.
Overfitting	When a model memorizes training data instead of learning generalizable patterns.
P Wave	The ECG component representing atrial depolarization (contraction of the upper chambers).
Precision	The proportion of predicted positives that are actually positive. Measures false alarm rate.
QRS Complex	The ECG component representing ventricular depolarization. The largest, sharpest deflection.
Recall	The proportion of actual positives correctly identified. Measures detection rate (sensitivity).
ReduceLROnPlateau	A callback that reduces the learning rate when validation loss stops improving.
ReLU	Rectified Linear Unit. Activation function: max(0, x). Simple, effective, solves vanishing gradient for positive values.
Residual Block	A building block containing convolutional layers with a skip connection.
ResNet	Residual Network. Architecture using skip connections to enable training of very deep networks.
R-peak	The highest point of the QRS complex. Used as the reference point for heartbeat segmentation.
SA Node	Sinoatrial Node. The heart's natural pacemaker, located in the right atrium.
Skip Connection	A shortcut that bypasses layers, adding the input directly to the output.
SMOTE	Synthetic Minority Over-sampling Technique. Creates new minority samples by interpolating between existing ones.
Softmax	Activation function that converts raw scores into a probability distribution summing to 1.
Support	The number of actual instances of each class in the dataset.
T Wave	The ECG component representing ventricular repolarization (electrical reset).
Validation Set	A subset of training data used to monitor performance during training, separate from the test set.

Vanishing Gradient	The problem where gradients become very small in deep networks, preventing effective learning.

Chapter 8

Quick Revision Notes

These condensed notes are designed to be read in 5-10 minutes immediately before a presentation or viva.

Project Summary

Goal: Classify ECG heartbeats into 5 AAMI classes using a 1-D ResNet.
Challenge: Extreme class imbalance (82.7% Normal beats).
Solution: SMOTE to balance training data.
Result: +4.67% recall improvement for Class S, +4.32% for Class F.

Key Numbers to Remember

Item	Value
Training Samples	87,554
Test Samples	21,892
Signal Length	187 points (1.5 sec at 125 Hz)
Number of Classes	5 (N, S, V, F, Q)
Class N Dominance	82.7%
Class F Rarity	0.7% (only 641 samples)
Baseline Accuracy	98.70%
SMOTE Accuracy	98.69%
Baseline Recall (S)	79.3%
SMOTE Recall (S)	84.0% (+4.67%)
Baseline Recall (F)	78.4%
SMOTE Recall (F)	82.7% (+4.32%)
Residual Blocks	5
Total Conv Layers	11
Optimizer	Adam (LR=0.001)
Loss	Sparse Categorical Crossentropy
Batch Size	256
Epochs	30 (with EarlyStopping)

Model Architecture Quick Reference

Architecture Summary

```
  Input(187,1) -> Conv1D(32,5)+ReLU
-> [ResBlock x5] -> Flatten
-> Dense(32)+ReLU -> Dense(32)+ReLU
-> Dense(5)+Softmax

Each ResBlock:
  Conv1D -> ReLU -> Conv1D -> Add(skip) -> ReLU -> MaxPool1D(5,2)
```

Key Definitions for Viva

Precision: Of predicted positives, how many are correct?

Recall: Of actual positives, how many did we find?

F1-Score: Harmonic mean of precision and recall

SMOTE: Creates synthetic minority samples via interpolation

ResNet: Uses skip connections to solve vanishing gradients

EarlyStopping: Stops training when validation loss plateaus

Fusion Beat: Two impulses (normal + ventricular) activate simultaneously

Supraventricular: Abnormal beat originating above the ventricles

Five-Second Answers

Why CNN? -- Automatically learns ECG features from raw signal

Why 1D? -- ECG is a 1D time series

Why ResNet? -- Skip connections prevent vanishing gradients

Why SMOTE? -- Balances classes without duplicating samples

Why accuracy is misleading? -- Dominated by 82.7% majority class

Why recall matters? -- Missing arrhythmias can be fatal

Main finding? -- SMOTE improved minority recall by ~5% with negligible accuracy loss

Final Takeaway

Important:
SMOTE improved minority heartbeat detection (Class S: +4.67%, Class F: +4.32%) while maintaining overall accuracy at 98.69%. The SMOTE model is clinically superior because it catches more dangerous arrhythmias, even though its mathematical accuracy is 0.01% lower.